



¹Institute of Marine Sciences, University of Chittagong, Chattogram-4331, Bangladesh

From four sampling sites along the Chattogram–Cox’s Bazar coast, 155 gastropod shells inhabited by hermit crabs were collected, with environmental parameters, temperature, salinity, pH, and dissolved oxygen (DO) recorded in situ. After morphological analysis, eleven species were identified, including six new distributional records for Bangladesh. Multivariate analyses (PCA and CCA) revealed salinity and temperature as key structuring factors, with *Coenobita* and *Diogenes* linked to high-salinity habitats at St. Martin’s Island, while *Clibanarius* dominated

estuarine and mangrove areas. Biodiversity indices and environmental parameter values are presented for each sampling site. An updated, annotated checklist integrating both previous records and newly identified taxa now documents fifteen species of hermit crabs in Bangladesh, belonging to four families and five genera.

Key words: Taxonomy, Biodiversity, Environmental, Morphology, Gastropod, South Coast

Hermit crabs, which belong to the Decapoda of the infra-order Anomura, are a diverse group of organisms found in the marine environment, while a small number of species can live in estuaries or even freshwater, which are hidden within empty gastropod shells. This study identified eleven species, assessed biodiversity indices across sites, and applied multivariate analyses (PCA and CCA) to examine relationships between habitats and environmental parameters. An updated annotated checklist was also compiled, integrating both previous records and newly identified species.

- 01 To identify hermit crabs based on their morphological characteristics
- 02 To record environmental parameters and analyze habitat associations of hermit crabs using PCA and CCA
- 03 To describe the biodiversity index of the hermit crabs of the Chattogram-iCox's Bazar coast
- 04 To provide an updated checklist of hermit crabs from the coastal regions of Bangladesh

Study Area

Cox's Bazar, St. Martin's Island

Sampling Locations

Nazirar Tek, Bakkhali River, Nunia Chora, St. Martin's Island

Morphometric Data

Total length-width
Carapace length-width
Cheliped length-width
Carpus length
Shape of rostrum

Meristic Data

Spine in carpus
Spine in cheliped
Teeth of dactyl

Statistical Data

Mean, Standard deviation, PCA, CCA, Biodiversity Index

Specimen Data

Identification up to Species Level & Show the Relationship between Habitat & Environmental Parameter

Output

Data Analysis

Figure 01 External morphology of Hermit crab

Figure 02 Study area map showing sampling locations

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Figure 17 consists of four bar charts arranged in a 2x2 grid, each showing the values of a different biodiversity index across five categories: Nunta Chora, Nazira Tek, Bakballi River estuary, St. Martin's Island, and an Average. The indices are Shannon (H), Simpson (1-D), Margalef (D), and Pielou's (J').

- Shannon (H):** The y-axis ranges from 0 to 2. The values are approximately: Nunta Chora (1.5), Nazira Tek (1.3), Bakballi River estuary (1.4), St. Martin's Island (1.7), and Average (1.4).
- Simpson (1-D):** The y-axis ranges from 0.66 to 0.94. The values are approximately: Nunta Chora (0.76), Nazira Tek (0.75), Bakballi River estuary (0.72), St. Martin's Island (0.91), and Average (0.76).
- Margalef (D):** The y-axis ranges from 0 to 2.5. The values are approximately: Nunta Chora (1.7), Nazira Tek (2.1), Bakballi River estuary (2.1), St. Martin's Island (1.9), and Average (1.9).
- Pielou's (J'):** The y-axis ranges from 0.25 to 1.05. The values are approximately: Nunta Chora (0.9), Nazira Tek (1.0), Bakballi River estuary (0.85), St. Martin's Island (0.9), and Average (0.9).

Genus/species	Habitat	ICUN status	Locality
<i>Clibanarius</i> (Dana, 1852)			
1. <i>Clibanarius infuspinatus</i>	Marine	LC	Cox's Bazar, St. Martin's
2. <i>Clibanarius padavensis</i>	Marine	NE	Cox's Bazar, St. Martin's
3. <i>Clibanarius longitarsus</i>	Brackish, Marine	LC	Sundarbans
4. <i>Clibanarius arehusa</i> *	Brackish, Marine	NE	Cox's Bazar, St. Martin's
5. <i>Clibanarius clibanarius</i>	Brackish, Marine	LC	Cox's Bazar, St. Martin's
<i>Coenobita</i> (Latreille, 1829)			
6. <i>Coenobita variabilis</i>	Terrestrial	DD	Sundarbans
7. <i>Coenobita violascens</i>	Marine	DD	Northern BOB
8. <i>Coenobita rugosus</i> *	Marine	NE	St. Martin's
9. <i>Coenobita perlatus</i>	Marine	NE	St. Martin's
<i>Diogenes</i> (Dana, 1852)			
10. <i>Diogenes planimanus</i> *	Marine	NE	St. Martin's
11. <i>Diogenes dubius</i> *	Marine	NE	St. Martin's
12. <i>Diogenes rectimanus</i>	Brackish, Marine	NE	Sonadia Island
<i>Pagurus</i> (Dana, 1852)			
13. <i>Pagurus bernhardus</i>	Brackish, Marine	DD	Cox's Bazar, St. Martin's
14. <i>Pagurus japonicus</i> *	Brackish, Marine	NE	Cox's Bazar,
<i>Parapagurus</i> (Smith, 1879)			
15. <i>Parapagurus nudus</i>	Marine	DD	Not Known

DD= Data Deficient; LC= Least Concern; NE =Not Evaluated, *New species

Family	Genus	Species
Diogenidae	<i>Clibanarius</i> (Dana, 1852)	<i>Clibanarius infraspinatus</i> (Hilgendorf, 1869)
		<i>Clibanarius padavensis</i> (De Man, 1888)
		<i>Clibanarius longitarsus</i> (De Haan, 1849)
		<i>Clibanarius arethusa</i> (De Mann, 1888)
		<i>Clibanarius clibanarius</i> (Herbst, 1791)
	<i>Diogenes</i> (Dana, 1852)	<i>Diogenes planimanus</i> (Henderson, 1893)
		<i>Diogenes dubius</i> (Herbst, 1804)
Coenobitidae	<i>Coenobita</i> (Latreille, 1829)	<i>Coenobita violascens</i> (Heller, 1862)
		<i>Coenobita rugosus</i> (H. Milne Edwards, 1837)
		<i>Coenobita perlatus</i> (H. Milne-Edwards, 1837)
Paguridae	<i>Pagurus</i> (Fabricius, 1775)	<i>Pagurus japonicus</i> (Stimpson, 1858)

Hermit crabs play an important ecological role in tropical marine ecosystems. Overall, this study enhances the baseline knowledge of hermit crab taxonomy and ecology along the Chattogram coast. The information generated will benefit scientists, conservation biologists, and policymakers in developing strategies for marine biodiversity conservation and sustainable coastal resource management.

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